CSE382 – Milestone 1 Progress Report

City Hotel (H2) Booking Demand – data preparation and baseline experiments

Repository: https://github.com/seifahmed2006/repo

Google Colab Link: https://colab.research.google.com/drive/1kiBcZf3zCePy4xihrz9Vqljh-aSs78mj?usp=sharing

Team:

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Dataset: Hotel Booking Demand, City Hotel (H2) – António, N., de Almeida, A., & Nunes, L. (2019), Data in Brief 22, 41–49.

All numbers below are produced by `python main.py milestone1` (seed 42). Tables/figures are in `outputs/`.

Only the **training** split (cross-validation) and the **validation** split were used. The test split was created once and never read.

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1. Completed work

- Repository structure, `config.yaml`, `requirements.txt`, CLI (`main.py`), README with exact commands, 11 pytest checks.
- H2.csv loaded and verified (79,330 × 31).
- EDA, data-quality audit, prediction point, leakage audit, reproducible split.
- Reusable preprocessing pipelines (fit on train only).
- One untuned baseline per task (+ a trivial reference model each).

1.1 Key results at a glance

- **Regression (ADR):** untuned Linear Regression reaches validation MAE 14.96, RMSE 21.03 and R^2 0.705, against MAE 30.29 for the mean predictor.
- **Classification (IsCanceled):** untuned Logistic Regression reaches validation accuracy 0.730 (majority class 0.699) and ROC-AUC 0.779, but recall is only 0.344, so most cancellations are missed.
- **Clustering:** untuned K-Means (k = 8) gives silhouette 0.162 (train) / 0.165 (validation) and Davies–Bouldin 1.381 / 1.362; among k ≥ 3 the sweep favours k = 5.
- **Leakage control:** 2 direct and 5 indirect post-booking fields are excluded, ADR is excluded for classification, and 25,902 duplicate rows are removed before the split. The test split is untouched.

2. Data understanding (outputs/eda/)

2.1 Loading and verification of H2.csv

The file was loaded unchanged and checked against the H2 specification before any cleaning:

- **Shape:** 79,330 rows × 31 columns, matching the published description of the City Hotel dataset.
- **Schema:** all 31 column names match the source labels in the project specification exactly. No column was renamed, so no label mapping is needed.
- **Arrival dates:** rebuilt from `ArrivalDateYear`, `ArrivalDateMonth` and `ArrivalDateDayOfMonth`; the range is 2015-07-01 to 2017-08-31, as documented for H2. (`ReservationStatusDate` spans 2014-10-17 to 2017-09-07; it is excluded from modelling, see §3.)
- **Dataset selection:** only H2.csv (City Hotel) is used. H1 (Resort Hotel) was not loaded or merged.

2.2 Data-quality findings and decisions

Finding	Value	Decision
Padded strings (`'HB '`) and `"NULL"` stored as text	Agent 8,131 (10.2%), **Company 75,641 (95.4%)**; Country 24, Children 4 are true NaN	Strip whitespace on load; convert `"NULL"` → missing.
Company 95% missing	–	Replaced by binary `HasCompany`.
Agent 10% missing, 223 IDs (224 incl. NULL)	–	Missing → category `Unknown`; rare IDs (<1% of train) grouped by the encoder.
Country 166 levels, Children 4 NaN	–	Country: `Unknown` + rare grouping. Children NaN → 0.
Exact duplicate rows	**25,902 (32.7%)**	Dropped **before** splitting (see §4).
Zero-guest bookings (Adults+Children+Babies = 0)	167 raw (154 after dedup)	Dropped (invalid).
ADR = 0	1,208	Kept for classification/clustering; **excluded from the regression task** (complimentary stays are not a price signal).
ADR max 5,400	1 row > 1000	Dropped (data-entry error).
Zero-night stays	331	Kept (flagged).
Skewed variables (abs. skew > 2)	Babies 44.8, PreviousBookingsNotCanceled 23.1, ADR 23.2, PreviousCancellations 16.6, DaysInWaitingList 10.3, BookingChanges 7.2 …	`log1p` for clustering; trees (Milestone 2) are insensitive; linear baselines are standardised.
Class balance (raw)	41.7% canceled	After de-duplication the **training** cancel rate is **30.1%** (see caveat in §4).

Figures (Appendix): target balance and missing values (Fig. 1), ADR distribution (Fig. 2), skewed variables (Fig. 3), cancellation rate by category (Fig. 4), lead time and seasonality (Fig. 5), correlations (Fig. 6), leakage evidence (Fig. 7).

2.3 Variable types, cardinality and ranges (raw file, 79,330 rows)

Variable	Type	Unique	Min / median / max	Skew
IsCanceled (target)	binary	2	0 / 0 / 1	–
LeadTime	integer	453	0 / 74 / 629	1.4
ArrivalDateYear / Month / WeekNumber / DayOfMonth	temporal	3 / 12 / 53 / 31	2015-07-01 to 2017-08-31	–
StaysInWeekendNights	integer	14	0 / 1 / 16	1.1
StaysInWeekNights	integer	29	0 / 2 / 41	3.0
Adults / Children / Babies	integer (Children: 4 NaN)	5 / 4 / 5	Adults 0 / 2 / 4; Children 0 / 0 / 3; Babies 0 / 0 / 10	−0.4 / 4.3 / 44.8
PreviousCancellations	integer	10	0 / 0 / 21	16.6
PreviousBookingsNotCanceled	integer	73	0 / 0 / 72	23.1
BookingChanges	integer	21	0 / 0 / 21	7.2
DaysInWaitingList	integer	115	0 / 0 / 391	10.3
ADR (regression target)	numeric	5,405	0 / 99.9 / 5,400	23.2
RequiredCarParkingSpaces	integer	4	0 / 0 / 3	6.3
TotalOfSpecialRequests	integer	6	0 / 0 / 5	1.4
IsRepeatedGuest	binary	2	0 / 0 / 1	–
Meal, MarketSegment, DistributionChannel, DepositType, CustomerType, ReservationStatus	categorical	4, 8, 5, 3, 4, 3	–	–
ReservedRoomType / AssignedRoomType	categorical	8 / 9	–	–
Country / Agent / Company	categorical (high cardinality)	166 / 223 / 207 (excl. NULL)	–	–
ReservationStatusDate	date	–	2014-10-17 to 2017-09-07	–

Skew values in Figure 3 differ from the table above because the figure is computed after cleaning (duplicates and invalid rows removed), whereas this table uses the raw file.

3. Prediction point, features, preliminary leakage audit

Prediction/analysis point for all three tasks: when the booking is first created. A field is allowed only if its value is fixed at that moment.

Direct leakage

Column	Evidence	Decision
`ReservationStatus`	Perfect mapping: Canceled/No-Show → IsCanceled=1 (33,102 rows), Check-Out → 0 (46,228)	Excluded everywhere
`ReservationStatusDate`	Date the final status was set = cancellation/check-out date	Excluded everywhere

Indirect leakage / timing (excluded because the value is set or changes after booking)

Column	Evidence (cancel rate by value, outputs/eda/leakage_cancel_rate_by_postbooking_fields.csv)
`RequiredCarParkingSpaces`	31.2% cancel at 0 spaces, **0.0% at ≥1** – no canceled booking has parking
`TotalOfSpecialRequests`	38.4% → 23.1% → 20.9% → 16.2% as requests rise; requests can be added after booking
`BookingChanges`	Accumulates until cancellation/check-in; outcome-dependent
`DaysInWaitingList`	Resolved after booking; outcome-correlated
`AssignedRoomType`	Assigned near check-in; cancel rate 33.2% when room matches vs **5.3% when it differs** (a mismatch can only exist for stays that happened)
`ADR` (classification only)	Final averaged rate, can be revised after booking; ablation planned for Milestone 2

Task-specific points

- **Regression (ADR):** the target is ADR. Inputs are the same booking-time fields as the classifier below: `LeadTime`, arrival date fields (`ArrivalDateYear`, `ArrivalDateMonth` as a categorical variable, week number, day of month, weekday), `StaysInWeekendNights`, `StaysInWeekNights`, `TotalNights`, `Adults`, `Children`, `Babies`, `TotalGuests`, `Meal`, `Country`, `MarketSegment`, `DistributionChannel`, `IsRepeatedGuest`, `PreviousCancellations`, `PreviousBookingsNotCanceled`, `ReservedRoomType`, `DepositType`, `Agent`, `CustomerType`, `HasCompany`. Excluded: `IsCanceled` and `ReservationStatus*` (outcomes), and every post-booking field listed above (`RequiredCarParkingSpaces`, `TotalOfSpecialRequests`, `BookingChanges`, `DaysInWaitingList`, `AssignedRoomType`). `ArrivalDateYear` is kept as a price-trend proxy – a caveat when predicting for future years.
- **Classification (IsCanceled):** inputs = lead time, arrival date fields, stay length, party size, meal, country, market segment, channel, repeat-guest history, reserved room, deposit type, agent, customer type, `HasCompany`.
- **Clustering:** only a small booking-profile set: `LeadTime, TotalNights, TotalGuests, ADR, PreviousCancellations, IsRepeatedGuest` + `MarketSegment, DepositType, CustomerType, Meal`. Excluded: outcome/post-booking fields above, and high-cardinality `Country/Agent/Company` (a 166-level one-hot would dominate Euclidean distance). ADR is kept here because it describes the booking (value segment), not an outcome; `IsCanceled` is used **only as an external check after clustering**, never to fit or choose parameters.
- **Clustering leakage risks:** (1) `ADR` is kept as a value-segment attribute. Unlike the classifier, the clusters are never fitted to `IsCanceled`, so ADR cannot leak the label into a supervised fit, but ADR = 0 is rarer among cancelled bookings, so a run without ADR is planned as an ablation. (2) `DepositType` is a booking-time field, but Non Refund bookings cancel 97% of the time in this data, which could make clusters partly mirror the outcome. It is kept for now and will be ablated together with ADR. In the k = 8 baseline, the most common deposit type in every cluster is No Deposit, so DepositType is not the dominant driver of the grouping. (3) `PreviousCancellations` and `IsRepeatedGuest` are booking-time and kept; their alignment with cancellation is reported as an external check only.

Leakage risks by task

Task	Main risks identified	Mitigation
Regression (ADR)	Outcome fields (`IsCanceled`, `ReservationStatus*`) recorded after the booking; post-booking fields (`AssignedRoomType`, `TotalOfSpecialRequests`, `BookingChanges`, `RequiredCarParkingSpaces`, `DaysInWaitingList`); duplicate rows landing in both train and test; target-based row filters; `ArrivalDateYear` as a trend proxy.	Booking-time feature set only; duplicates removed before the split; filters are fixed rules applied to all splits; year caveat documented.
Classification (IsCanceled)	Direct leakage (`ReservationStatus`, `ReservationStatusDate`); indirect leakage (parking, special requests, booking changes, waiting list, assigned room); ADR, which can be revised after booking; `DepositType` (Non Refund is 97% cancelled); duplicates across splits.	All listed fields excluded; ADR ablation and DepositType ablation planned; deduplication before the stratified split.
Clustering	Outcome or post-booking fields shaping the groups; `ADR` and `DepositType` partly mirroring the outcome; `PreviousCancellations` aligning with cancellation; scaler or encoder fitted on data that includes the evaluation rows.	Only a booking-profile feature set; `IsCanceled` used as an external check after clustering, never to fit or choose parameters; ADR and DepositType ablations planned; preprocessing fitted on the training split.

4. Split strategy and preprocessing

1. Load → strip/NULL fix → drop duplicates, zero-guest, ADR>1000 (**deterministic filters only, no fitted statistics**) → 53,273 rows.
2. Stratified (on `IsCanceled`) split, seed 42: **train 37,291 / validation 7,991 / test 7,991** (70/15/15).
3. Regression additionally keeps only ADR>0: train 36,643 / val 7,836.
4. Pipeline per task: numeric → median impute → standardise; categorical → `Unknown` impute → one-hot (categories <1% of train grouped). Fitted **inside** `Pipeline`, so cross-validation refits it per fold. Clustering adds `log1p` on skewed counts and ADR capped at 500 (constants from config).
5. Feature engineering is stateless (`TotalNights`, `TotalGuests`, `HasCompany`, `ArrivalWeekday`).

Why de-duplicate: the file has no booking ID, so some "duplicates" are genuine separate bookings. But an identical row in train and validation/test makes the score partly a lookup. In an un-deduplicated random split we measured that ~39% of validation rows had an exact (features + label) twin in train. **Caveat:** de-duplication lowers the cancel rate (41.7% → 30.1%), so results describe *distinct* booking profiles. A sensitivity run on the non-deduplicated data is planned for Milestone 2.

ADR data-quality filters. The rules ADR ≤ 1000 (all tasks) and ADR > 0 (regression only; ADR = 0 rows are kept for classification and clustering) use the value of the regression target to drop rows. They are fixed, rule-based data-quality filters applied identically to the train, validation and test splits; no statistic is learned from the data. Test rows are filtered but never scored or inspected in Milestone 1.

4.1 Separation of stages (leakage control)

- **Cleaning:** deterministic and row-wise (whitespace, "NULL", duplicates, zero guests, ADR > 1000). Nothing is fitted, and it runs before the split.
- **Feature engineering:** stateless functions (`TotalNights`, `TotalGuests`, `HasCompany`, `ArrivalWeekday`) that use only the row itself; no target or cross-row statistics.
- **Preprocessing:** imputers, scalers and one-hot encoders (including the rare-category grouping) are fitted inside a scikit-learn Pipeline on training data only. During cross-validation they are refitted on each training fold.

5. Initial results (untuned baselines)

5.1 Experiment protocol

All baselines share one protocol: same split (seed 42), 5-fold cross-validation on the training split for model comparison, and one fit on the full training split scored on the validation split. Test data is not used.

Task / model	Features and preprocessing	Hyperparameters	Evaluation	Fit time (s)	# features
Regression: mean predictor (reference)	–	predicts the training mean	CV + validation	–	–
Regression: Linear Regression	booking-time features (§3); median impute, standardise numerics, one-hot categoricals (<1% grouped)	untuned defaults	CV + validation	0.72	81
Classification: majority class (reference)	–	predicts the majority class	CV + validation	–	–
Classification: Logistic Regression	same as regression (no ADR, no post-booking fields)	untuned defaults; no class weights; seed 42	CV + validation	1.52	–
Clustering: K-Means	LeadTime, TotalNights, ADR (log1p, capped at 500), TotalGuests, IsRepeatedGuest, PreviousCancellations; one-hot MarketSegment, DepositType, CustomerType, Meal; standardised	k = 8 (sklearn default); seed 42	silhouette and Davies–Bouldin on train and validation	1.78	–

Regression – ADR (ADR>0)

Model	CV MAE (5-fold, train)	Val MAE	Val RMSE	Val R^2
Mean predictor (reference)	30.63	30.29	38.71	−0.000
Linear Regression	**14.91 ± 0.20**	**14.96**	**21.03**	**0.705**

Cross-validation (5-fold, training split): Linear Regression MAE 14.91 ± 0.20, RMSE 21.19, R^2 0.714; mean predictor RMSE 39.64, R^2 −0.000. Fit time 0.72 s with 81 encoded features.

Interpretation: linear regression halves the error of the naive baseline and explains ~70% of ADR variance from booking-time fields alone (season, room type, market segment, party size). CV and validation agree, so there is no sign of overfitting. Errors are largest at the price extremes (validation, by ADR quartile): the most expensive quartile (ADR > 133) is under-predicted by ≈ 15 on average (MAE 20.2) and the cheapest quartile is over-predicted by ≈ 9 (MAE 15.1) – the usual shrinkage of a linear model toward the mean. September has the highest monthly MAE (21.0). This suggests non-linear models (Milestone 2) should help, which we will test rather than assume.

Classification – IsCanceled

Model	CV Acc	CV F1	CV ROC-AUC	Val Acc	Val Prec	Val Rec	Val F1	Val ROC-AUC	Val PR-AUC
Majority class (reference)	0.699	0.000	0.500	0.699	0.000	0.000	0.000	0.500	0.301
Logistic Regression	0.731	0.432	0.780	0.730	0.586	0.344	0.434	0.779	0.581

Confusion matrix (validation, threshold 0.5, 7,991 bookings):

	Predicted: not canceled	Predicted: canceled
Actual: not canceled	5,002 (TN)	584 (FP)
Actual: canceled	1,577 (FN)	828 (TP)

Of the 2,405 cancellations in validation, 1,577 (66%) are missed; of the 1,412 bookings flagged as canceled, 828 (59%) really are. This matches the reported recall (0.344) and precision (0.586).

Interpretation: accuracy is only slightly above the majority baseline (0.730 vs 0.699), so accuracy alone is misleading. ROC-AUC 0.78 shows real ranking signal, but **recall is low (0.34)** at the default 0.5 threshold – most cancellations are missed. Planned: class weights/threshold tuning, non-linear models. The largest coefficients are `DepositType` (Non Refund ↑), specific countries (PRT ↑) and agents – consistent with the risks in §3.

Clustering – K-Means (untuned: k = 8, sklearn default)

Config	Train silhouette	Train Davies–Bouldin	Val silhouette	Val Davies–Bouldin
k = 8	0.162	1.381	0.165	1.362

k-sweep (train, outputs/clustering/kmeans_k_sweep.csv): among k ≥ 3 the best silhouette/DB is **k = 5** (0.216 / 1.287). k = 2 has silhouette 0.62 but only because it isolates 1,147 repeat guests (3% of rows) – a trivial split, not a useful segmentation. Structure is weak overall (silhouette ≈ 0.2), typical for mixed booking data.

Calinski–Harabasz index for k = 8: 7,131 (train) and 1,549 (validation); the two differ partly because the index grows with the number of rows. K-Means fit time: 1.78 s.

Baseline cluster profile (k = 8; kmeans_baseline_cluster_profile.csv) – examples:

- Cluster 1 (6,309): last-minute bookings (≈ 5 days lead time), short stays (1.7 nights), Online TA.
- Cluster 0 (5,690): Offline TA/TO, long lead time (≈ 119 days), `Transient-Party` customers.
- Cluster 6 (9,039): long stays (≈ 5.4 nights), booked ≈ 97 days ahead.
- Cluster 4 (5,406): ≈ 3.2 guests per booking, highest ADR (≈ 170) – family/premium bookings.
- Clusters 3 & 5 (807 / 336): Corporate **repeat guests**, low ADR (≈ 67); cluster 5 also has many previous cancellations (mean 2.5).
- Cluster 7 (9,237): typical online booking, ≈ 2.3 nights, ≈ 85 days lead time.
- Cluster 2 (467): long lead time (≈ 139 days) **and previous cancellations (mean 1.0); 99.8% were canceled** (external check only).

Clusters differ in booking behaviour, and some align with cancellation, which is expected but must not be used to tune k.

Known limitations

- **De-duplication changes the population:** the cancel rate drops from 41.7% to 30.1%, so results describe distinct booking profiles. A sensitivity run on the non-deduplicated data is planned.
- **Random split ignores time:** bookings span 2015–2017, so a time-based hold-out would be a stricter test.
- **Only linear baselines so far:** the expected gain from non-linear models is a hypothesis, not yet a result.
- **Weak cluster structure:** silhouette is about 0.2 for k ≥ 3, and k = 2 is a trivial split (repeat guests).
- **Borderline features:** `ArrivalDateYear`, `DepositType` and ADR are flagged in §3 and will be ablated rather than assumed safe.

6. Member contributions

Member	GitHub account	Work completed
Seif eldin Ahmed Saeed Hanafy Ibrahim	seifahmed2006	Repository setup and EDA module
Peter Hany Youssef Sedek Mina	peter199h	Preprocessing pipelines and tests
Hamza Karim Kashmiry	HamzaKashmiryy	Regression and classification baselines
Cherif Chadi Georges Antoun Ibrahim	chichochago2109-lgtm	Clustering baseline and report

7. Reproducibility

- **Environment:** `python -m venv .venv`, activate it, then `pip install -r requirements.txt`. Python 3.13.18, scikit-learn 1.6.1, pandas 2.2.3 (also numpy 2.4.6, matplotlib 3.10.0, PyYAML 6.0.3), verified on Google Colab.
- **Data:** place `H2.csv` (City Hotel) in `data/` or run `python main.py download`; the file is verified on load (§2.1).
- **Run everything:** `python main.py milestone1` regenerates all tables and figures in `outputs/` (seed 42, parameters in `config.yaml`).
- **Tests:** `python -m pytest -q` runs the 11 checks.
- **Artifacts:** following the submission rules, no fitted model or pipeline files are included in the ZIP; the commands above regenerate them locally.

Appendix – Figures

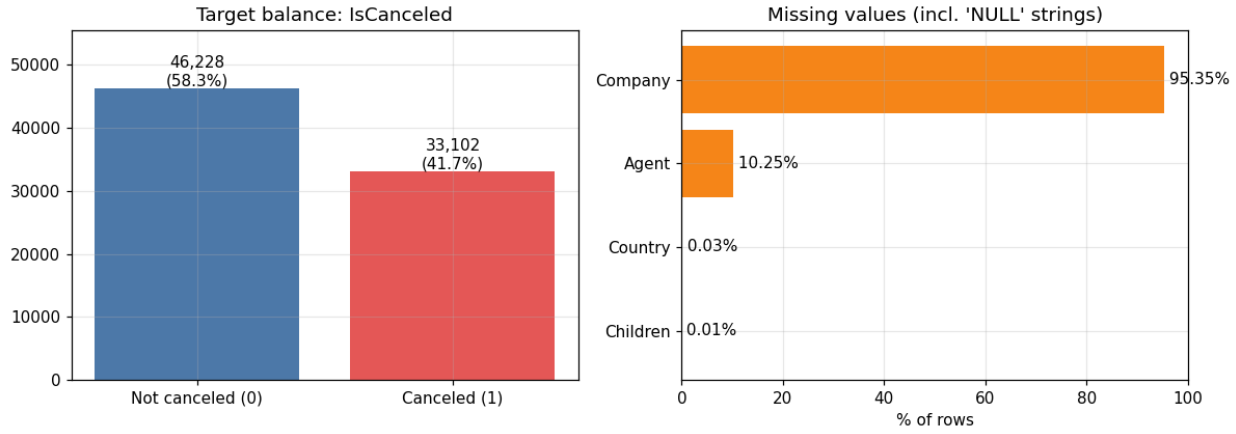


Figure 1. Target balance and missing values (incl. 'NULL' strings).

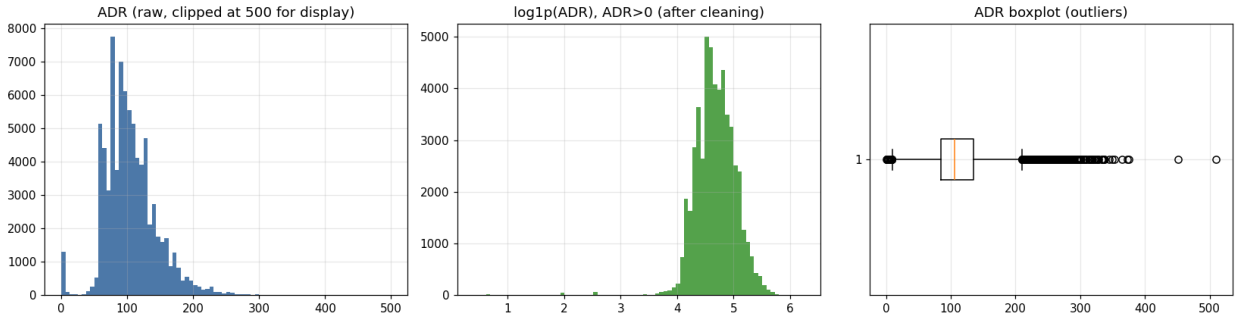


Figure 2. ADR distribution: raw, log-scale and boxplot.

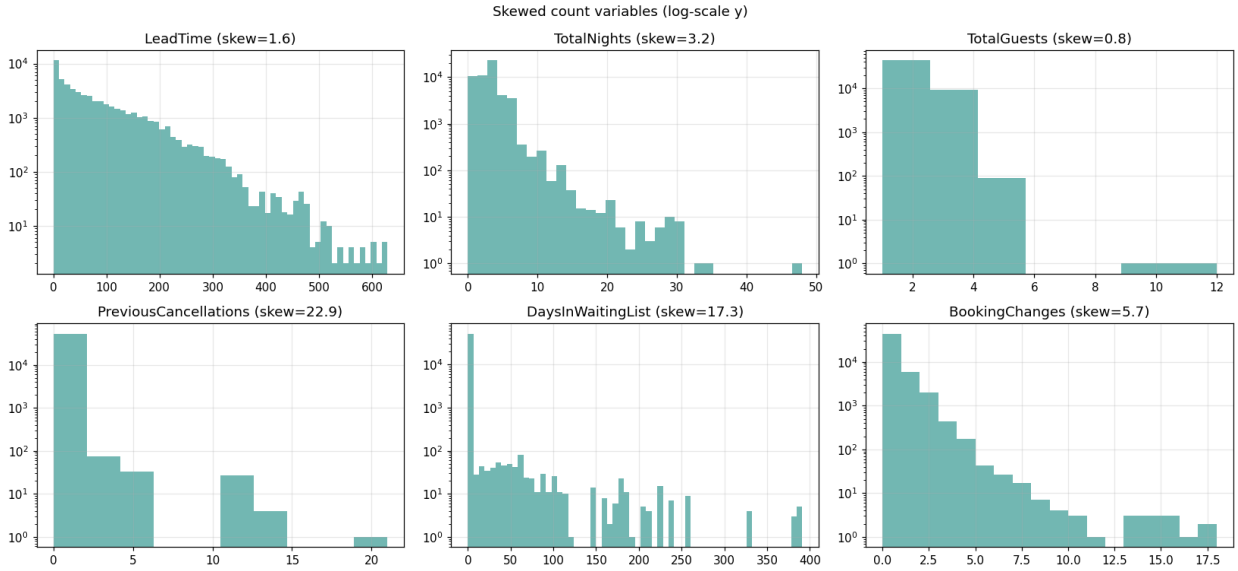


Figure 3. Skewed count variables (log-scale y-axis; computed after cleaning).

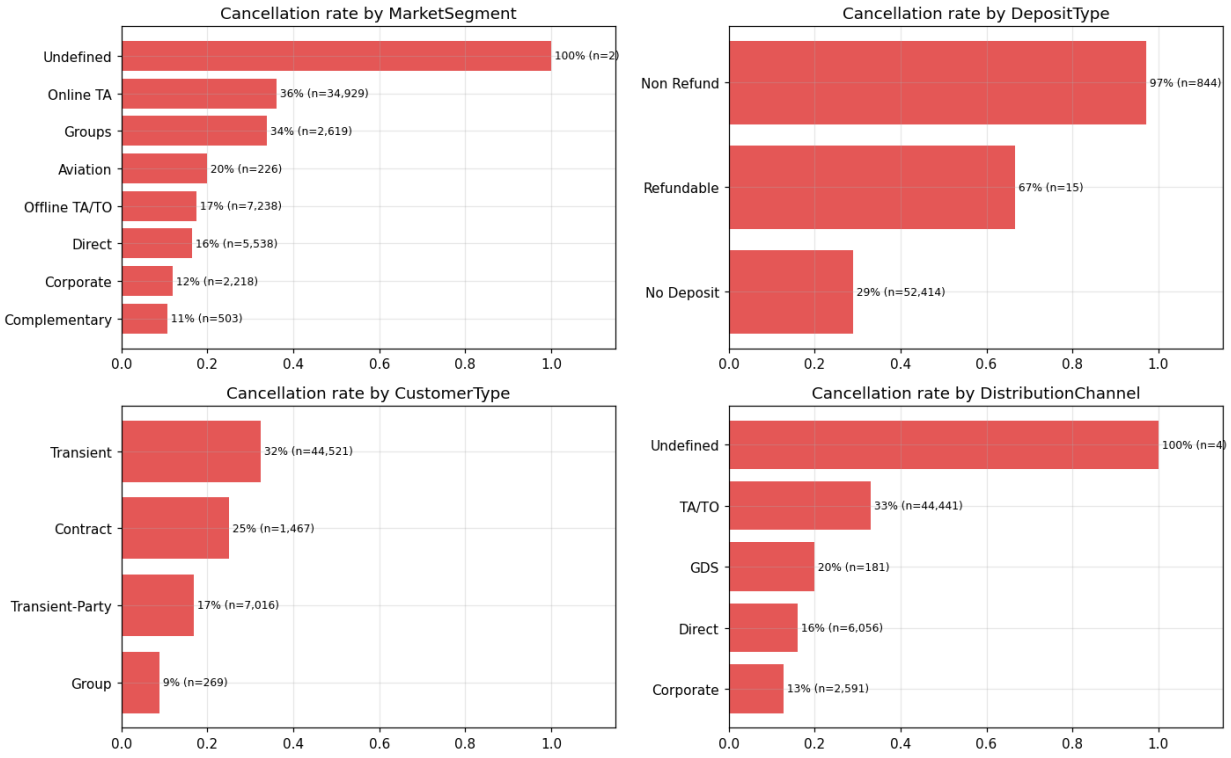


Figure 4. Cancellation rate by market segment, deposit type, customer type and channel.

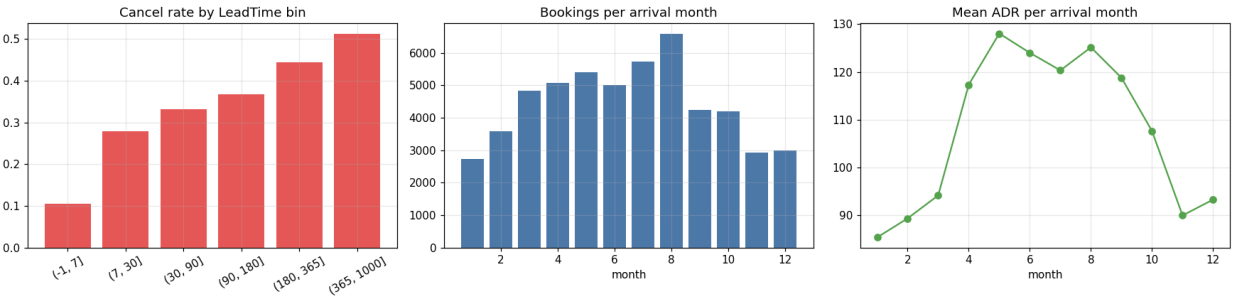


Figure 5. Lead time vs cancellation, and seasonality of bookings and ADR.

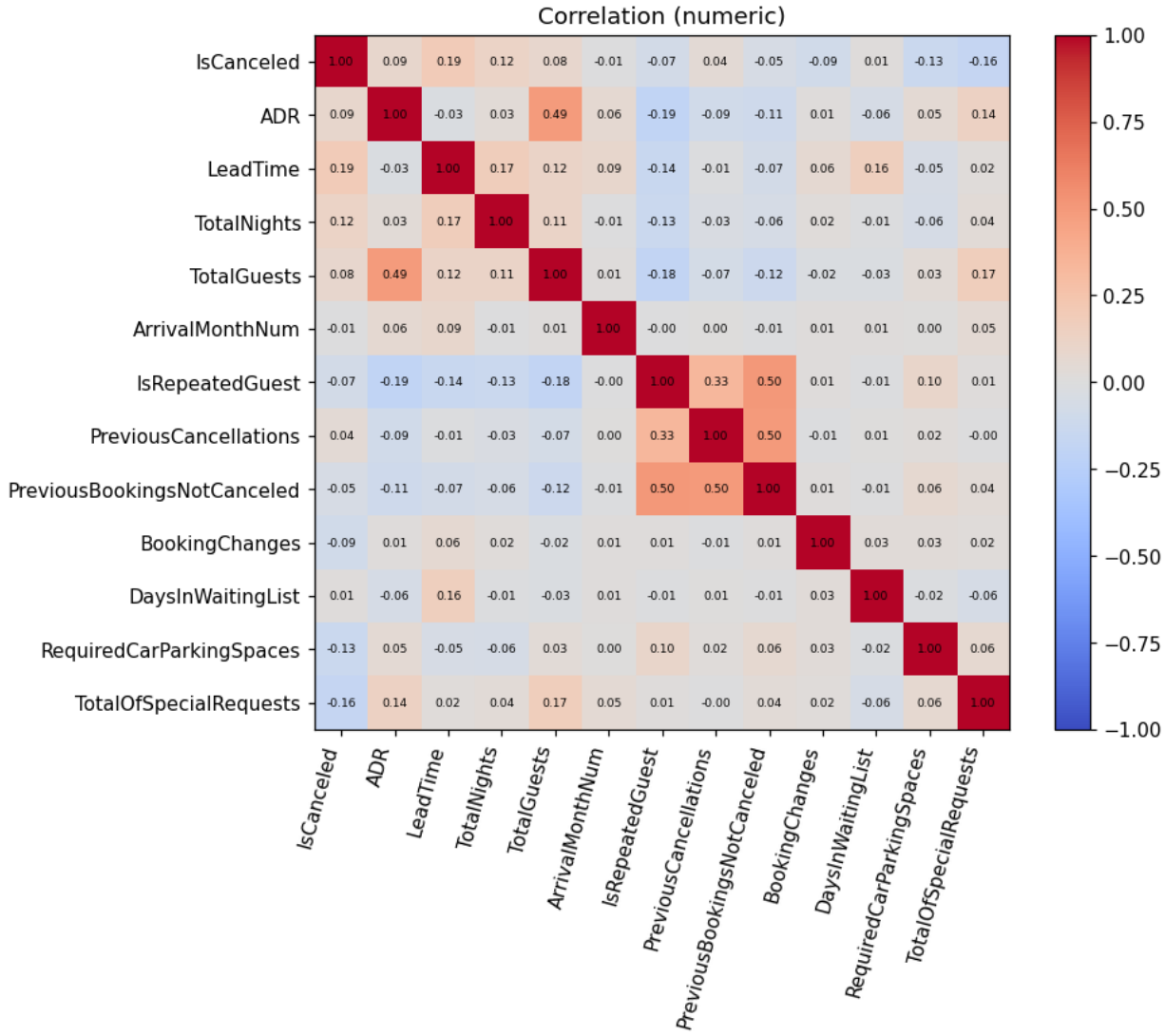


Figure 6. Correlation matrix of numeric variables.

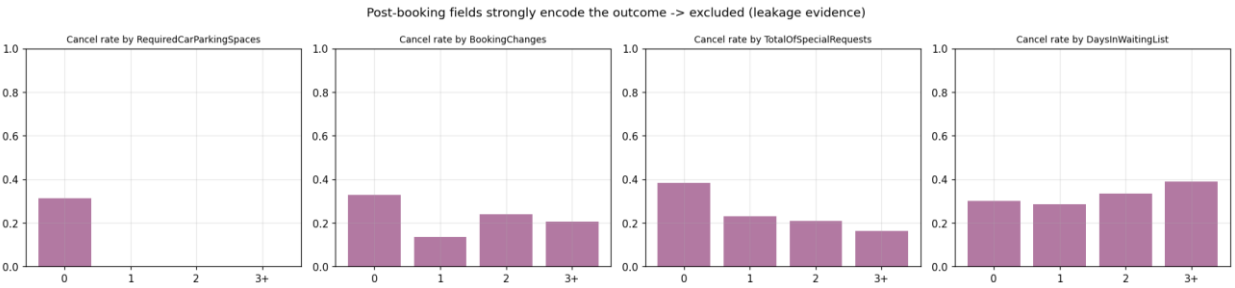


Figure 7. Leakage evidence: post-booking fields strongly encode the outcome.

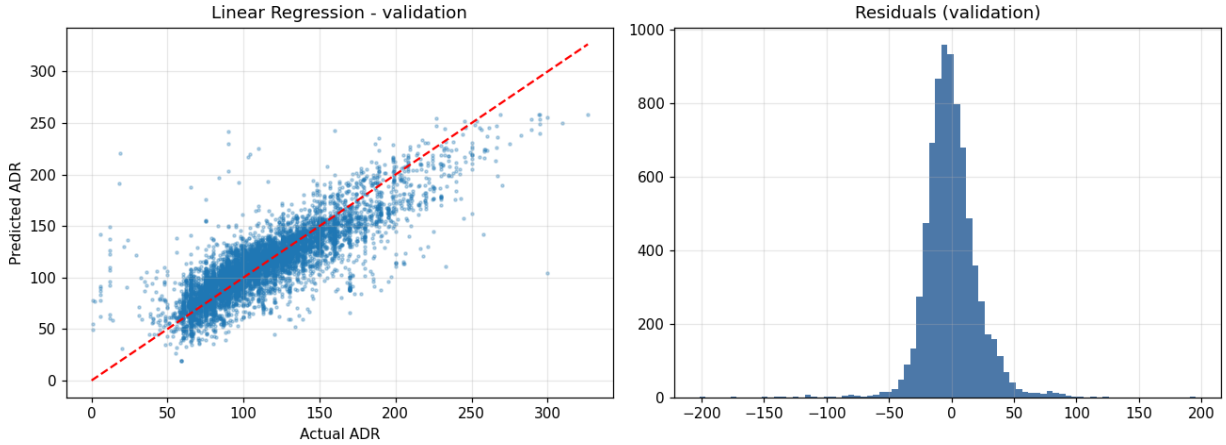


Figure 8. Linear Regression baseline: predicted vs actual ADR and residuals (validation).

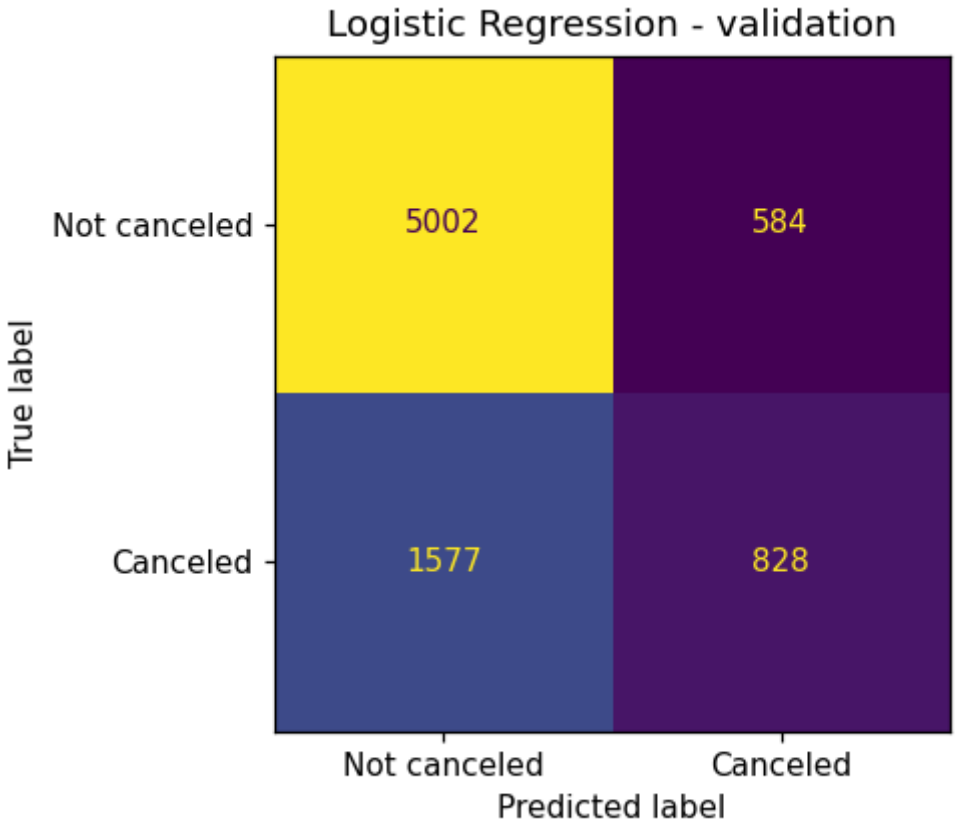


Figure 9. Logistic Regression baseline: confusion matrix (validation).

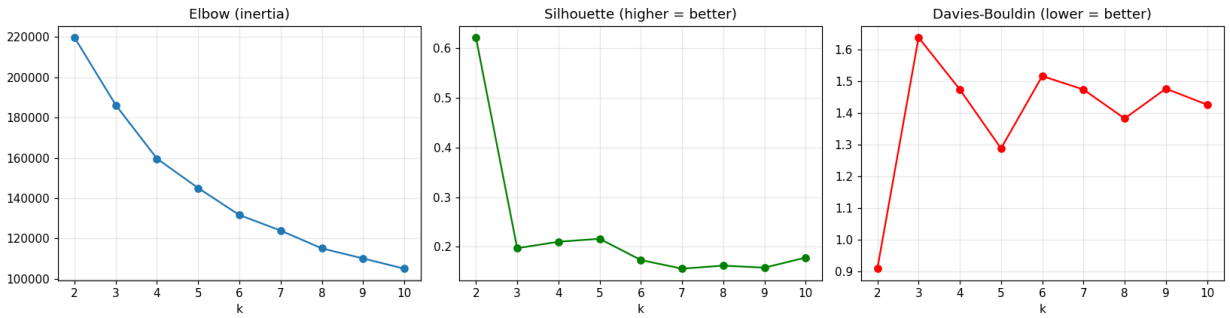


Figure 10. K-Means k-sweep: inertia, silhouette and Davies–Bouldin (train).

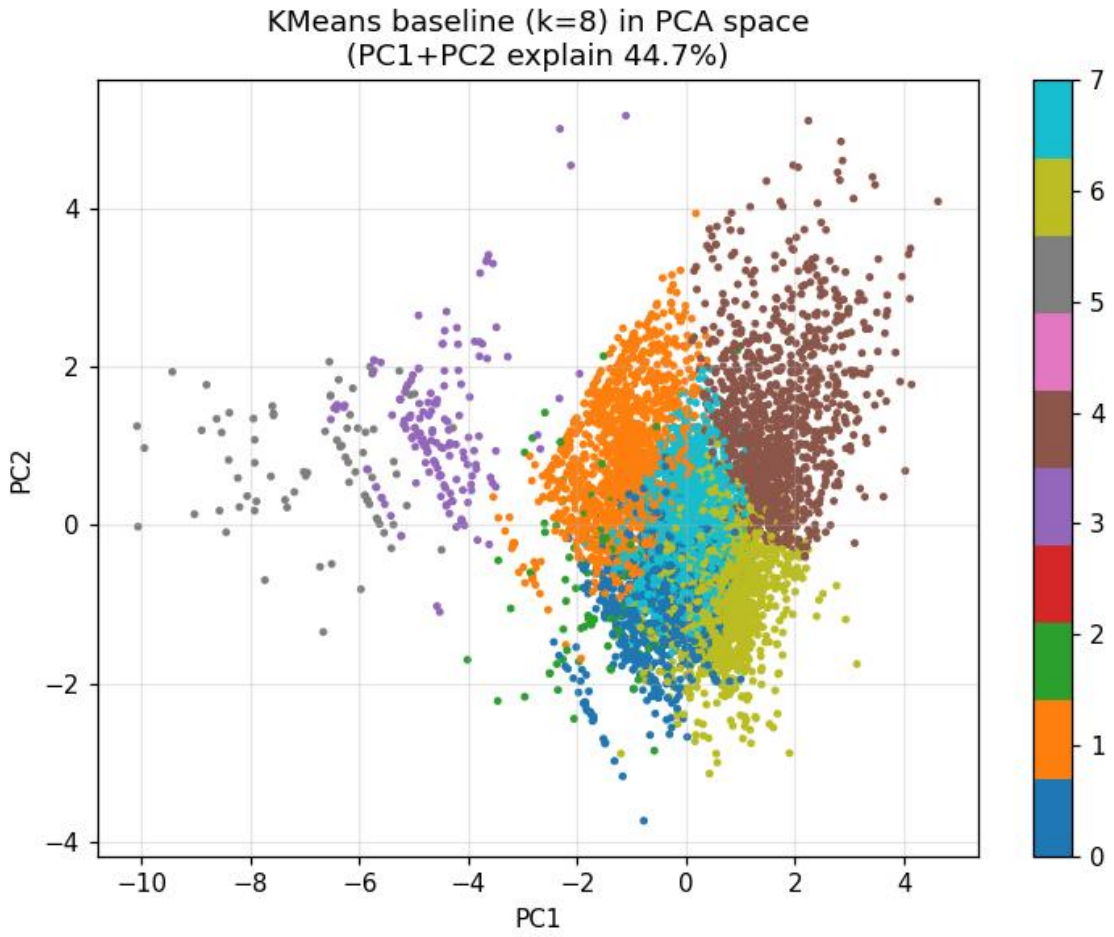


Figure 11. K-Means baseline (k = 8) projected onto the first two principal components.